

Chapter

Nitric Acid

1. Diammonium hydrogen phosphate, $(\text{NH}_4)_2 \text{HPO}_4$, is a widely used fertiliser. Which of the following substance could be added to this compound to produce a mixture that provides the three principle elements needed for plant growth?

COMP 2023

- (a) Magnesium nitrate
- (b) Ammonium sulphate
- (c) Ammonium phosphate
- (d) Potassium chloride

2. Material used to slow down the speed of gases in absorption tower.

MAIN 2023

- (a) Platinum
- (b) Gold
- (c) Silver
- (d) Quartz

3. What poisons the catalytic chamber ? MAIN 2022

- (a) Nitrogen monoxide
- (b) Impurities
- (c) Platinum guaze
- (d) Excess air

4. Concentration of nitric acid obtained in Ostwald's process. SQP 2021

- (a) 10%
- (b) 5%
- (c) 15%
- (d) 20%

5. On photochemical decomposition of HNO_3 , the products formed are
SQP 2022

- (a) NO_2 and O_2
- (b) NO and O
- (c) N_2O_4 and NO_2
- (d) N_2O_5 and NO_2

6. Chloride formed when nitric acid and hydrochloric acid reacts together.
COMP 2021

- (a) Ammonium chloride
- (b) Hydrochloric acid
- (c) Nitrosyl Chloride
- (d) Ammonia

7. Hard crust formed when chile salt petre and sulphuric acid reacts together.
MAIN 2021

- (a) Na_2SO_4
- (b) Na_2CO_3
- (c) NaSiO_3
- (d) NaCl

8. A nitrate which on heating leaves no residue behind. MAIN 2022

9. Name two metals which react with very dilute nitric acid. SQP 2023

10. Nitric acid is a _____, _____ acid. COMP 2022

11. The salts of nitric acid are called _____. MAIN 2023

12. Nitric acid in laboratory is prepared by the reaction of _____ and _____ acid.
SQP 2021

13. Heating of ammonium nitrate (name only the nitrogen containing compound) COMP 2022

14. Write an equation for the following reaction : Action of heat on potassium nitrate. SQP 2022

15. Sodium nitrate and concentrated sulphuric acid . MAIN 2023
16. Nitre reacts with concentrated Sulphuric acid. SQP 2023
17. Copper oxide and dilute nitric acid. COMP 2023
18. Action of heat on AgNO_3 . SQP 2022
19. Write a balanced chemical equation for the preparation of nitric acid from potassium nitrate. COMP 2023
20. NaNO_3 gives NO_2 and O_2 on heating. SQP 2023
21. A nitrate which on heating leaves no residue behind. MAIN 2021
22. Give reasons for the following : The apparatus for the preparation of nitric acid by heating concentrated sulphuric acid and potassium nitrate, should be all glass. SQP 2021
23. During thunderstorm, rainwater contains nitric acid. Explain with reactions. MAIN 2023
24. Abhishek was given a salt 'X' which was white in colour for analysis. On strong heating it produced a yellow residue, a colourless gas and also a reddish-brown gas. The solution of the salt 'X' when tested with excess of ammonium hydroxide produced a chalky white insoluble precipitate.
- (i) Name the coloured gas evolved when Abhishek heated the salt strongly.
- (ii) Which cation was present in the sample given to Abhishek?
- (iii) Identify the salt given to Abhishek for analysis. SQP 2025
25. A nitrate which forms a precipitate with ammonium hydroxide and is also soluble in excess of it: SQP 2025
- (a) ferrous nitrate
- (b) ferric nitrate
- (c) lead nitrate
- (d) copper nitrate

Solution

1. (d) Potassium chloride

Diammonium hydrogen phosphate ($(\text{NH}_4)_2\text{HPO}_4$) provides **nitrogen (N)** and **phosphorus (P)** for plants. To supply the third essential element, **potassium (K)**, **potassium chloride (KCl)** is added. This combination gives a fertiliser containing **all three principal elements (N, P, K)** required for plant growth.

2. (d) Quartz

In an **absorption tower**, **quartz** is used to **slow down the speed of gases**, allowing better contact with the liquid absorbent. This improves the **efficiency of absorption**.

3. (b) Impurities

The **catalytic chamber** can be **poisoned by impurities** present in the reactants or exhaust gases. These impurities **reduce the effectiveness of the catalyst** and hinder the catalytic reaction.

4. (b) 5%

In the **Ostwald process**, the nitric acid obtained directly from the absorption of nitrogen oxides in water is **dilute**, having a **concentration of about 5%**.

5. Thus (a) is correct option

6. (c) Nitrosyl chloride (NOCl)

When **nitric acid (HNO_3)** and **concentrated hydrochloric acid (HCl)** react together, they produce **nitrosyl chloride (NOCl)** along with water. This reaction is used in the preparation of nitrosyl chloride in the laboratory.

7. Thus (a) is correct option.

8. Ammonium nitrate

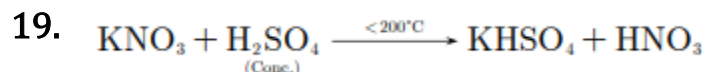
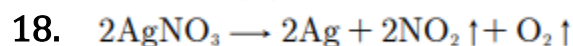
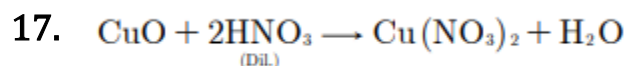
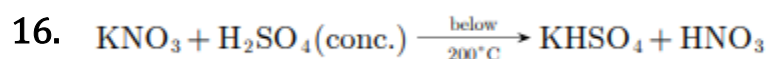
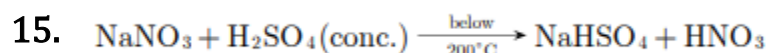
9. Magnesium and manganese.

10. strong, mono-basic

11. nitrates

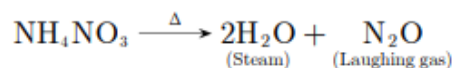
12. Potassium nitrate, concentrated Sulphuric

13. Nitrous oxide (N₂ O).



20. NaNO₃ gives O₂ on heating.

21. Ammonium nitrate (NH₄NO₃)



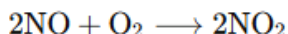
22. The apparatus for preparing **nitric acid** by heating **concentrated sulphuric acid** and **potassium nitrate** should be made of **glass** because **nitric acid vapours are highly corrosive**. They would **attack rubber, cork, or metals**, but **glass is resistant** to these vapours and remains unaffected.

23. During a **thunderstorm**, the high temperature of **lightning** (~3000°C) causes **nitrogen (N₂)** in the air to react with **oxygen (O₂)** to

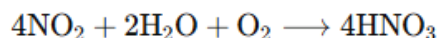
form **nitric oxide (NO)**:



The nitric oxide is then oxidised by atmospheric oxygen to form **nitrogen dioxide (NO₂)**:



The nitrogen dioxide dissolves in rainwater, and in the presence of oxygen, forms **nitric acid (HNO₃)**:



Hence, rainwater contains nitric acid during thunderstorms.

24. (i) Coloured gas evolved: The reddish-brown gas evolved on strong heating is **nitrogen dioxide (NO₂)**.

(ii) Cation present: The cation present in the salt is **lead (Pb²⁺)**.

(iii) Identity of the salt: The salt given to Abhishek is **lead nitrate (Pb(NO₃)₂)**, which decomposes on heating to give **PbO (yellow residue)**, **NO₂ (reddish-brown gas)**, and **O₂ (colourless gas)**.

25. (d) Copper nitrate

Copper nitrate (Cu(NO₃)₂) reacts with **ammonium hydroxide (NH₄OH)** to form a **bluish precipitate of copper hydroxide [Cu(OH)₂]**, which dissolves in excess **NH₄OH** to form a **deep blue complex [Cu(NH₃)₄]²⁺**.